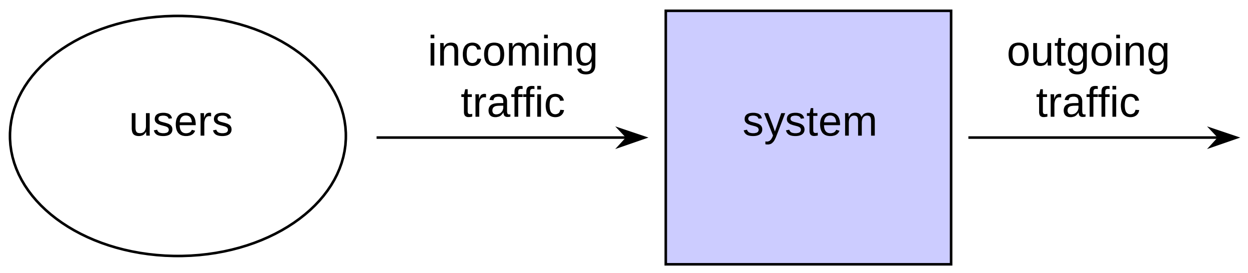
Teletraffic

- Purpose of Teletraffic Theory
- Teletraffic models
- Classical model for telephone traffic
- Classical model for data traffic

Traffic point of view

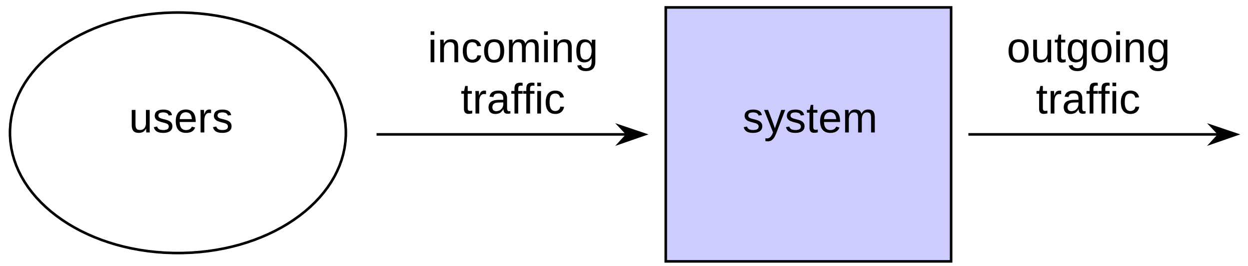
- Telecommunication system from the **traffic point of view**:



- Ideas:
  - the **system serves** the incoming **traffic**
  - the traffic is generated by the **users** of the system

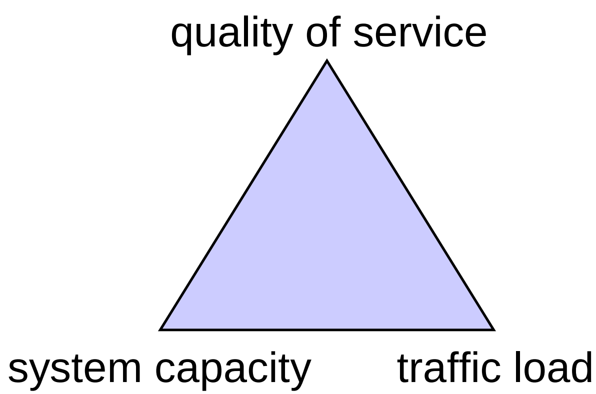
Interesting questions

- Given the system and incoming traffic,
  what is the quality of service experienced by the user?
- Given the incoming traffic and required quality of service,
  how should the system be dimensioned?
- Given the system and required quality of service,
  what is the maximum traffic load?



General purpose (1)

- Determine relationships between the following three factors:
  - **quality of service**
  - **system capacity**
  - **traffic load**



General purpose (2)

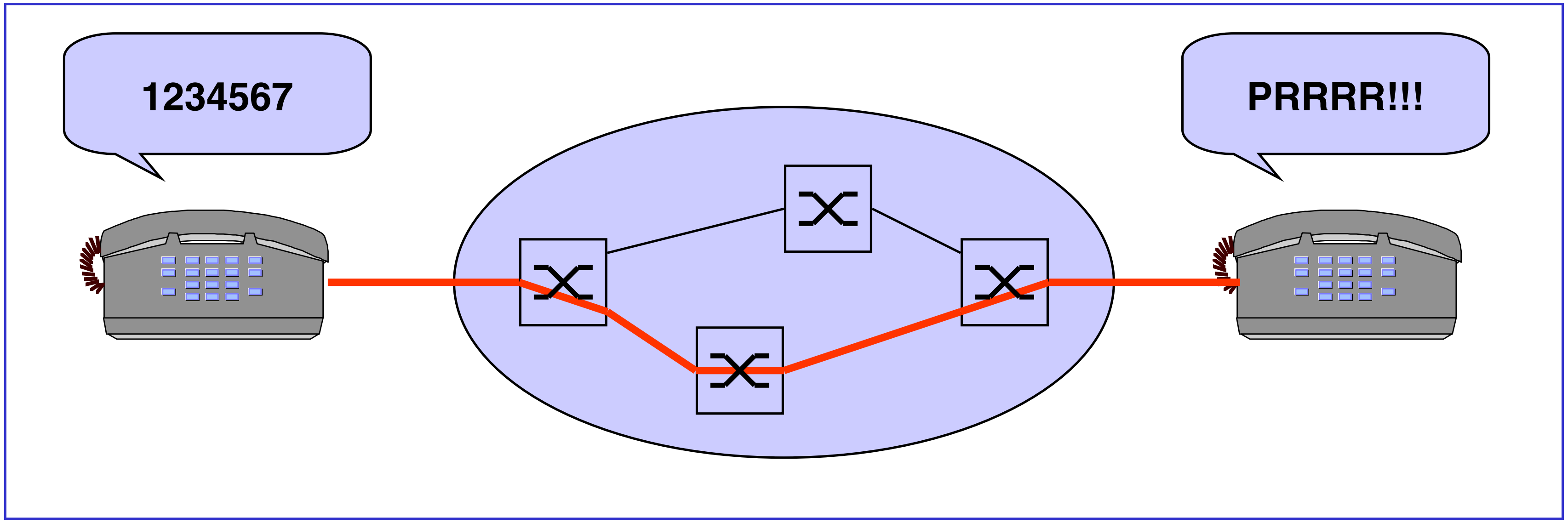
- System can be
  - a single device (e.g. a link between two telephone exchanges, a processor routing packets in a data network, a statistical multiplexer in ATM network) or
  - a whole communications network (e.g. telephone or data network) or a part of it
- Typically a system consists of
  - a device (hardware) and
  - principles controlling it (software)
- Traffic, on the other hand, consists of (depending on the case)
  - calls, packets, bursts, cells, etc.

General purpose (3)

- Quality of service can be described
  - from the users point of view
    - e.g. call blocking probability, distribution of delay experienced by a packet stream
  - from the system point of view (= **performance** of the system)
    - e.g. utilization
- It is also possible to think quality of service
  - from the offered traffic point of view
    - e.g. blocking experienced by ATM-connection requests, or other connection level property; grade of service
  - from the carried traffic point of view
    - e.g. cell loss during an accepted ATM connection, or other criteria during an accepted connection; quality of service

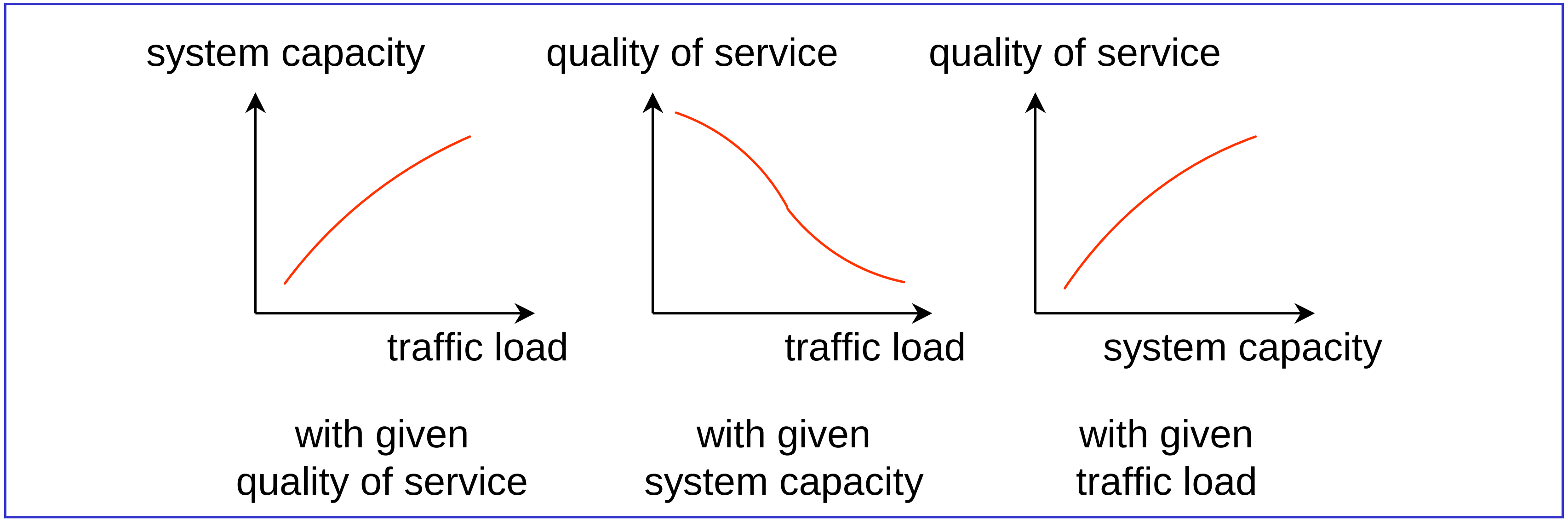
Example

- Telephone traffic
  - traffic = telephone calls
  - system = telephone network
  - quality of service = probability that “the line is not busy”



Relationships between the three factors

- **Qualitatively**, the relationships are as follows:



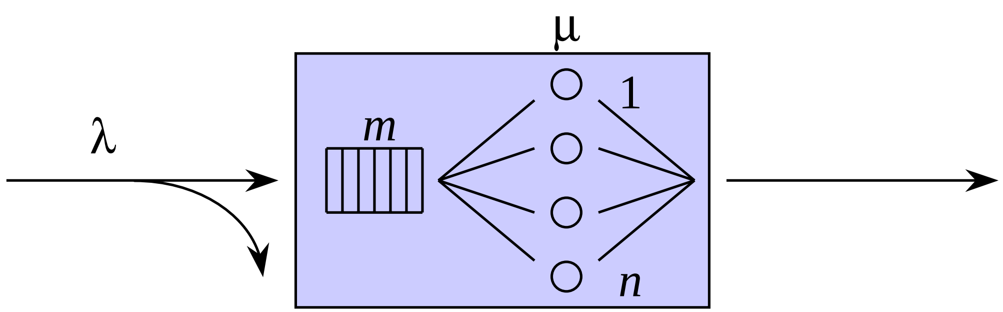
- To describe the relationships **quantitatively**, **mathematical models** are needed

Teletraffic models

- Two phases of teletraffic modelling:
  - modelling of the incoming traffic $\Rightarrow$ **traffic model**
  - modelling of the system itself $\Rightarrow$ **system model**
- Roughly, teletraffic models can be divided into two categories by the system model
  - **loss systems** (loss models)
  - **waiting/queueing systems** (queuing models)
- During this course, we present simple teletraffic models describing a single resource
- These models can be combined to create models for whole telecommunication networks
  - loss networks
  - queueing networks

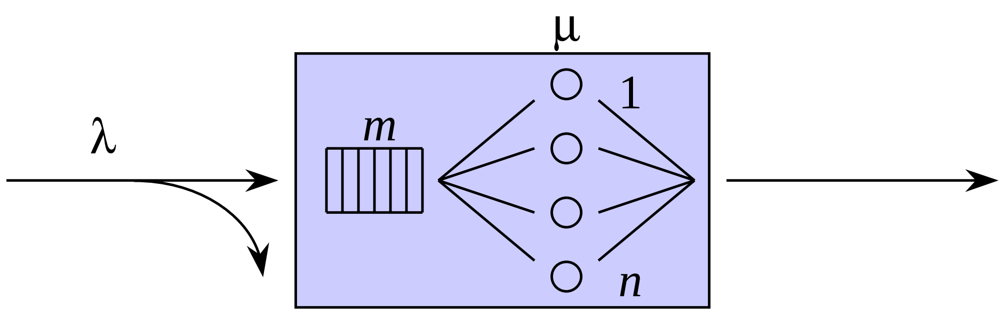
Simple teletraffic model

- **Customers arrive** at rate λ (customers per time unit on average)
  - $1/\lambda$ = average inter-arrival time
- Customers are **served** by n parallel **servers**
- When busy, a server serves at rate μ (customers per time unit)
  - $1/\mu$ = average service time of a customer
- There are m **waiting** places in the system
- It is assumed that blocked customers (arriving in a **full** system) are lost



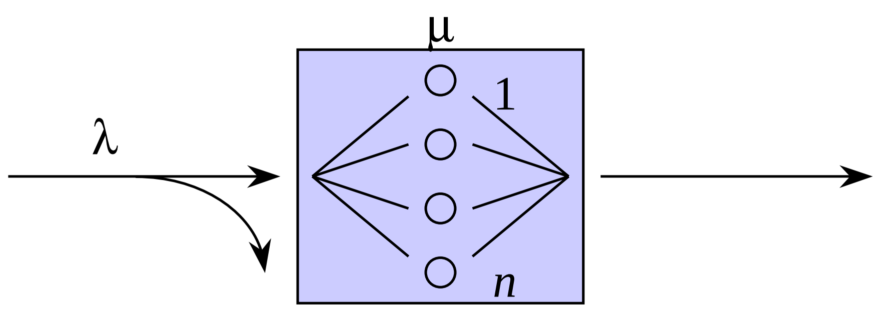
Exercise

- Consider the simple teletraffic model presented on the previous slide
  - What is the **traffic model**?
  - What is the **system model**?



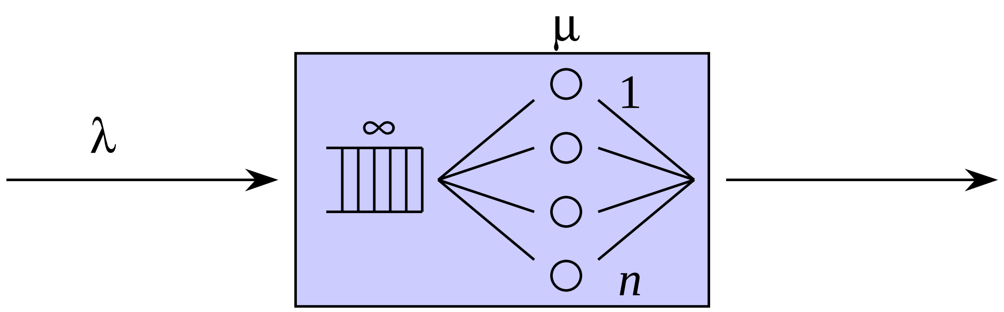
Pure loss system

- No waiting places ($m = 0$)
  - If the system is **full** (with all n servers occupied) when a customer arrives, she is not served at all but lost. System is thus **lossy.**
- From the customer's point of view, it is interesting to know e.g.
  - What is the probability that the system is full when she arrives?
- From the system's point of view, it is interesting to know e.g.
  - What is the utilization factor of the servers?



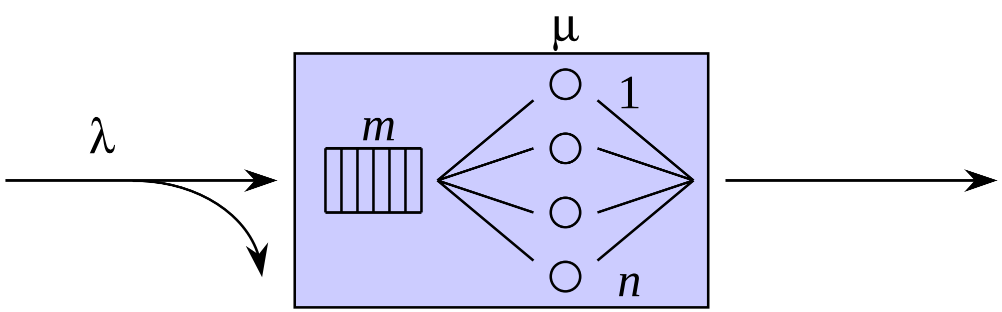
Pure waiting system

- Infinite number of waiting places ($m = \infty$)
  - If all n servers are occupied when a customer arrives, she occupies one of the waiting places. No customers are lost but some of them have to wait before getting served; system is thus **lossless.**
- From the customer's point of view, it is interesting to know e.g.
  - what is the probability that she has to wait "too long"?
- From the system's point of view, it is interesting to know e.g.
  - what is the utilization factor of the servers?



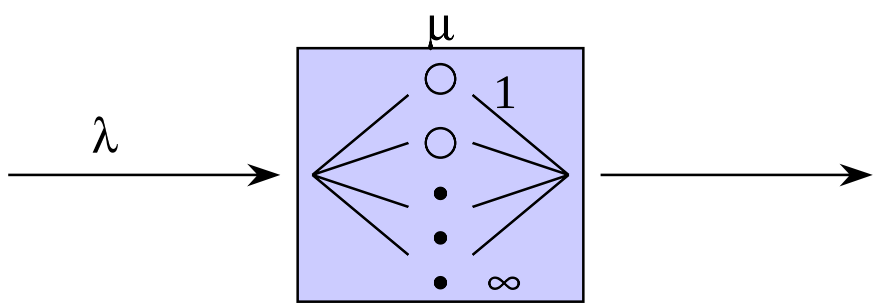
Mixed system

- Finite number of waiting places ($0 < m < \infty$)
  - If all n servers are occupied but there are free waiting places when a customer arrives, she occupies one of the waiting places
  - If all n servers and all m waiting places are occupied when a customer arrives, she is not served at all but lost
  - Some customers are lost and some customers have to wait before getting served. Also this system is thus lossy.

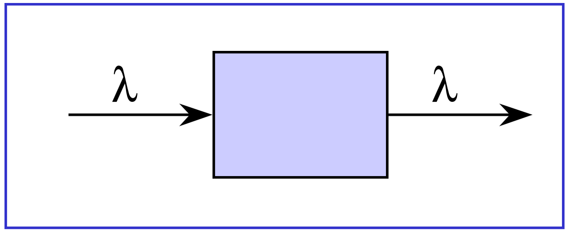


Infinite system

- Infinite number of servers ($n = \infty$)
  - No customers are lost or even have to wait before getting served; it is a lossless system.
  - This (hypothetical) system is often much more simple to analyze than the corresponding real system with finite capacity.
  - Sometimes, this can be the only way to get even some approximate results for the real system



Little's formula



- Consider a system where
  - new customers arrive at rate λ
- **Stability assumption**:
  - Customers do not accumulate in the system, occasionally system is empty
- Consequence:
  - Customers depart from the system at rate λ
- Denote

$$\overline{N} = \text{average number of customers in the system}$$

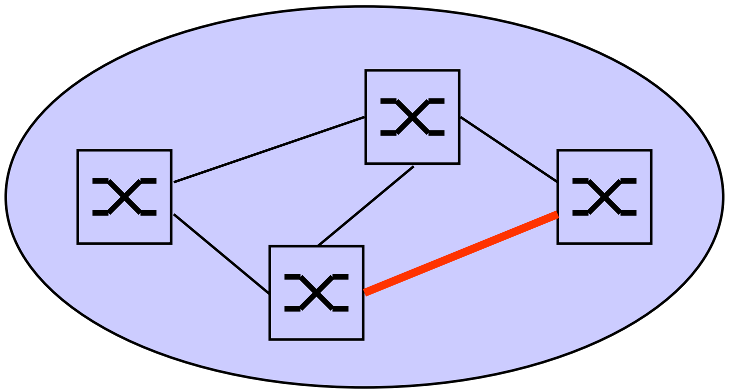
$$\overline{T} = \text{average time a customer spends in the system}$$

- **Little's formula**:

$$\overline{N} = \lambda \overline{T}$$

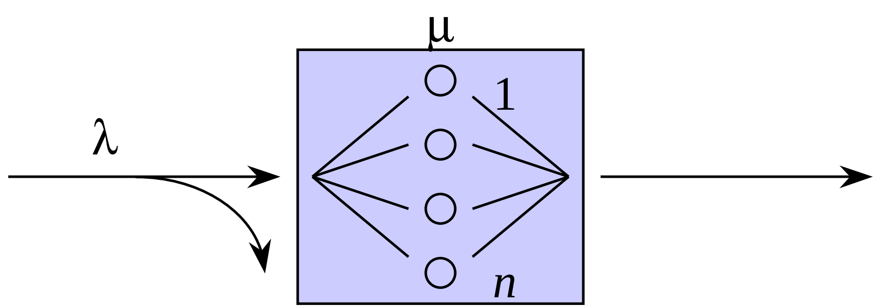
Classical model for telephone traffic (1)

- Loss models have traditionally been used to describe (circuit-switched) telephone networks
  - Pioneering work made by Danish mathematician A.K. Erlang (1878-1929)
- Consider a link between two telephone exchanges (classical teletraffic problem)
  - traffic consists of the ongoing telephone calls on the link

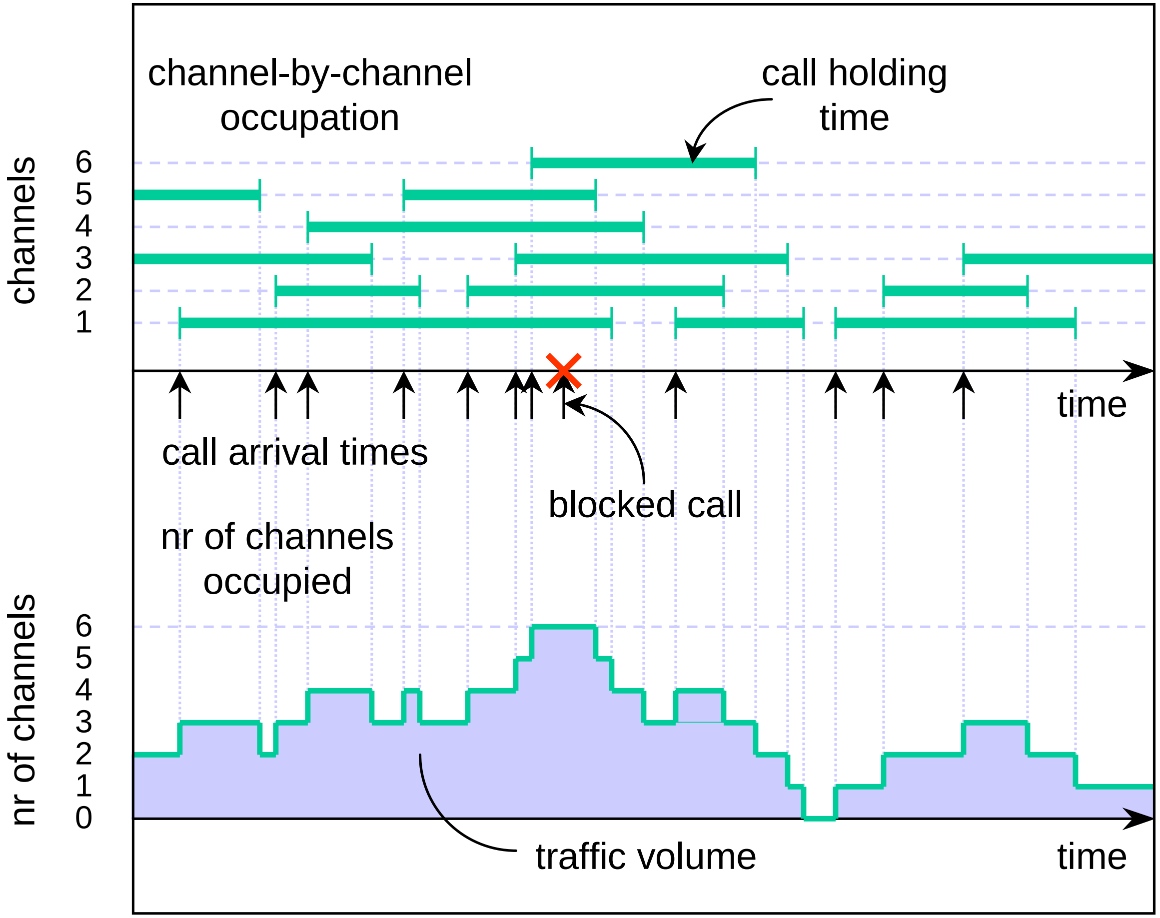


Classical model for telephone traffic (2)

- Erlang modelled this as a **pure loss system** ($m = 0$)
  - customer = call
    - λ = call arrival rate
  - service time = (call) holding time
    - $h = 1/\mu$ = average holding time
  - server = channel on the link
    - n = number of channels on the link



Traffic process



Traffic intensity

- In telephone networks:

Traffic $\leftrightarrow$ Calls

- The amount of traffic is described by the traffic intensity a
- **Definition:** the **traffic intensity** a is
  the product of the arrival rate λ and the mean holding time h:

$$a = \lambda h$$

- Note that the traffic intensity is a dimensionless quantity. However, to emphasize the context, the “unit” of the traffic intensity a is called **erlang** (**erl**)
- By Little’s formula: the traffic intensity tells the number of ongoing calls in the corresponding (hypothetical) infinite system.

Example

- Consider a local exchange. Assume that,
  - on the average, there are 1800 new calls in an hour, and
  - the mean holding time is 3 minutes
- It follows that the traffic intensity is

$$a = 1800 * 3/60 = 90 \text{ erlang}$$

- If the mean holding time increases from 3 minutes to 10 minutes, then

$$a = 1800 * 10/60 = 300 \text{ erlang}$$

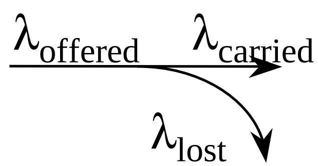
Blocking

- In a loss system some calls are lost
  - a call is lost if all n channels are occupied when the call arrives
  - the term **blocking** refers to this event
- There are (at least) two different types of blocking quantities:
  - **Call blocking** B_c = probability that an arriving call finds all n channels occupied = the fraction of calls that are lost
  - **Time blocking** B_t = probability that all n channels are occupied at an arbitrary time = the fraction of time that all n channels are occupied
- The two blocking quantities are not necessarily equal
  - If calls arrive according to a Poisson process, then $B_c = B_t$
- Call blocking is a better measure for the quality of service experienced by the subscribers which measured by grade of service (GoS)
- Typically, time blocking is easier to calculate

$$B_c = \frac{\text{Number of calls lost}}{\text{Number of calls offered}} = \frac{\text{Traffic lost}}{\text{Traffic offered}}$$

Call rates

- In a loss system, there are three types of call rates:
  - λ_{offered} = arrival rate of all call attempts
  - λ_{carried} = arrival rate of carried calls
  - λ_{lost} = arrival rate of lost calls



- Note:

$$\lambda_{\text{offered}} = \lambda_{\text{carried}} + \lambda_{\text{lost}} = \lambda$$

$$\lambda_{\text{carried}} = \lambda(1 - B_{\text{c}})$$

$$\lambda_{\text{lost}} = \lambda B_{\text{c}}$$

Traffic streams

- The three call rates lead to the following three traffic concepts:
  - **Traffic offered** $a_{\text{offered}} = \lambda_{\text{offered}} h$
  - **Traffic carried** $a_{\text{carried}} = \lambda_{\text{carried}} h$
  - **Traffic lost** $a_{\text{lost}} = \lambda_{\text{lost}} h$
- Note:

$$a_{\text{offered}} = a_{\text{carried}} + a_{\text{lost}} = a$$

$$a_{\text{carried}} = a(1 - B_{\text{c}})$$

$$a_{\text{lost}} = aB_{\text{c}}$$

- Traffic offered and traffic lost are hypothetical quantities, but traffic carried is measurable; by Little's formula, it is the average number of occupied channels on the link

Teletraffic analysis

- System capacity
  - n = number of channels on the link
- Traffic load
  - a = (offered) traffic intensity
- Quality of service (from the subscribers' point of view)
  - B_c = probability that an arriving call finds all n channels occupied
- If we assume an **M/G/*n*/*n* pure loss system**, that is
  - calls arrive according to a **Poisson process** (with rate λ)
  - call holding times are independently and identically distributed according to **any distribution** with mean h

Erlang's formula

- Then the quantitive relation between the three factors (system capacity, traffic load and quality of service) is given by the **Erlang's formula**

$$B_{\mathrm{c}} = \mathrm{Erl}(n,a) = \frac{\frac{a^n}{n!}}{\sum_{i=0}^{n} \frac{a^i}{i!}}$$

- Note:

$$n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1, \qquad 0! = 1$$

- Other names:
  - Erlang's blocking formula
  - Erlang's B-formula
  - Erlang's loss formula
  - Erlang's first formula

Example

- Assume that there are n = 4 channels on a link and the offered traffic is a = 2.0 erlang. Then the call blocking probability B_c is

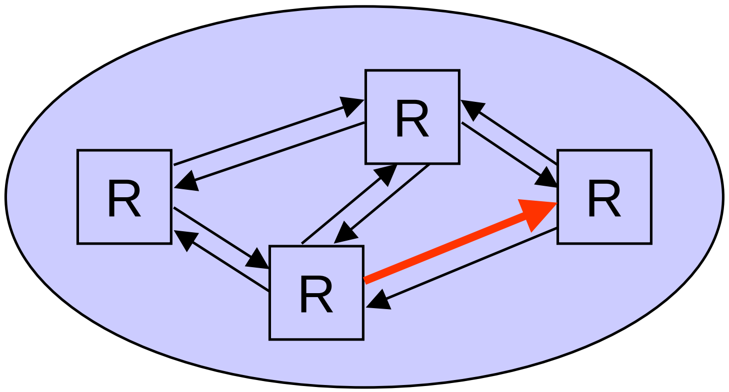
$$B_c = \mathrm{Erl}(4,2) = \frac{\frac{2^4}{4!}}{1+2+\frac{2^2}{2!}+\frac{2^3}{3!}+\frac{2^4}{4!}} = \frac{\frac{16}{24}}{1+2+\frac{4}{2}+\frac{8}{6}+\frac{16}{24}} = \frac{2}{21} \approx 9.5\%$$

- If the link capacity is raised to n = 6 channels, then B_c reduces to

$$B_c = \mathrm{Erl}(6,2) = \frac{\frac{2^6}{6!}}{1+2+\frac{2^2}{2!}+\frac{2^3}{3!}+\frac{2^4}{4!}+\frac{2^5}{5!}+\frac{2^6}{6!}} \approx 1.2\%$$

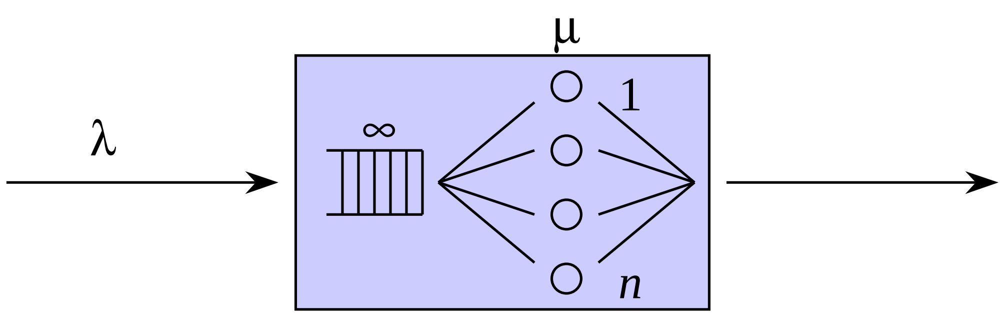
Classical model for data traffic (1)

- Queueing models are suitable for describing (packet-switched) data networks
- Consider a link between two packet routers
  - traffic consists of data packets transmitted on the link

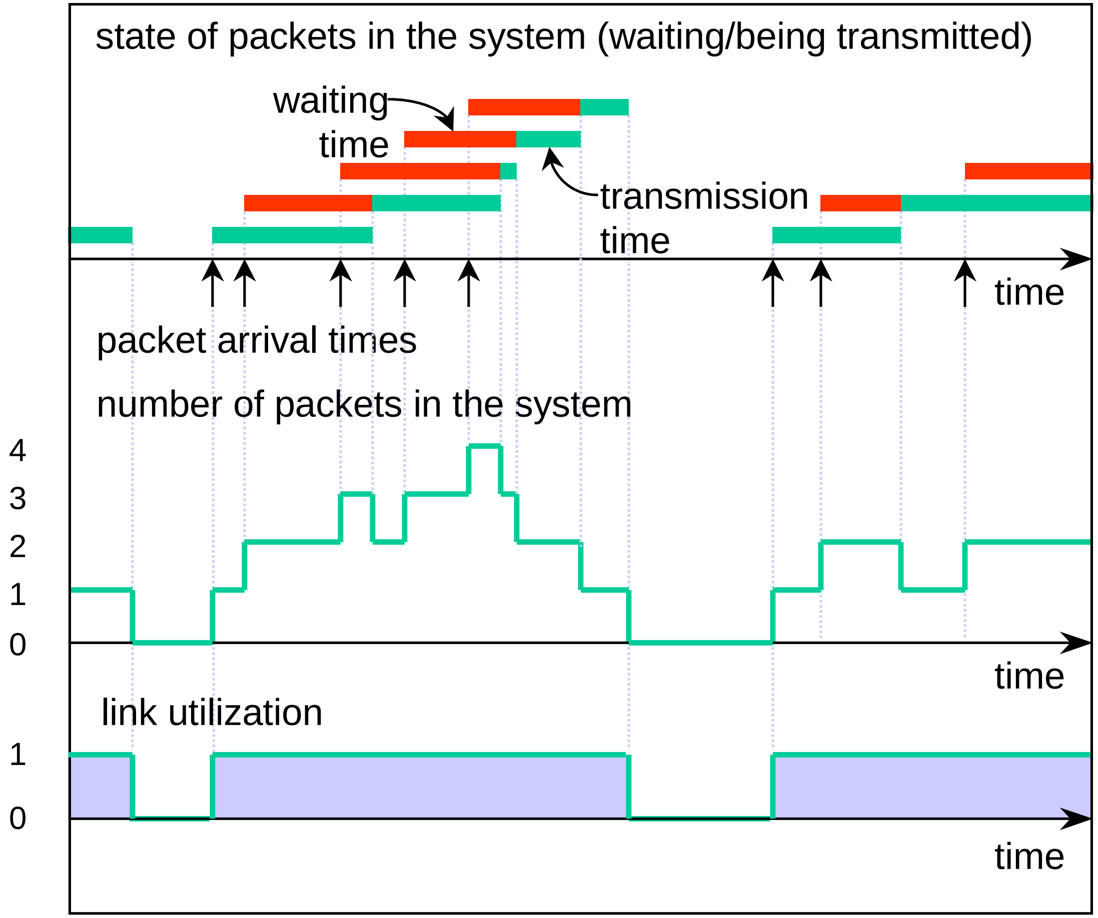


Classical model for data traffic (2)

- This can be modelled as a **pure waiting system** with a single server ($n = 1$) and an infinite buffer ($m = \infty$)
  - customer = packet
    - λ = packet arrival rate (packets per time unit)
    - L = average packet length (data units)
  - server = link, waiting places = buffer
    - C = link's speed (data units per time unit)
  - service time = packet transmission time
    - $1/\mu = L/C$ = average packet transmission time



Traffic process



Traffic load

- In packet-switched data networks:

Traffic $\leftrightarrow$ Packets

- The amount of traffic is described by the traffic load ρ
- **Definition:** the **traffic load** ρ is
  the quotient between the arrival rate λ and the service rate $\mu = C/L$:

$$\rho = \frac{\lambda}{\mu} = \frac{\lambda L}{C}$$

  - Note that the traffic load is a dimensionless quantity (as traffic intensity in loss systems)
  - By Little's formula:traffic load tells the number of customers in service = probability of "server busy" = the **utilization factor** of the server

Example

- Consider a link between two packet routers. Assume that,
  - on the average, 10 new packets arrive in a second,
  - the mean packet length is 400 bytes, and
  - the link speed is 64 kbps.
- It follows that the traffic load is

$$\rho = 10 * 400 * 8 / 64{,}000 = 0.5 = 50\%$$

- If the link speed is increased to 150 Mbps, then the load is just

$$\rho = 10 * 400 * 8 / 150{,}000{,}000 = 0.0002 = 0.02\%$$

- Note:
  - 1 byte = 8 bits
  - 1 kbps = 1 kbit/s = 1 kbit per second = 1,000 bits per second
  - 1 Mbps = 1 Mbit/s = 1 Mbit per second = 1,000,000 bits per second